

Ethan Yang

e99yang@uwaterloo.ca | 778-798-0213 | Richmond, BC | [Linkedin](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Bachelor of Science, Honours Mathematical Physics
GPA: 3.6

Waterloo, ON
Expected May 2030

EXPERIENCE

Western Pacific Robotics Academy (WPRA)

Robotics Instructor

Richmond, BC

Sept 2023 – Present

- Administer workshop logistics and maintenance, maintain accurate inventory records, and provide front-line customer service to parents and students.
- Instruct students on core VEX robotics concepts, including CAD design, C++ programming, and control system troubleshooting.
- Facilitate the academy's growth as an instructor, achieving a 60%+ increase in active membership.

Team Captain & Mechanical Lead – VEX Teams 604X and 886W

Sept 2019 – May 2025

- Led the mechanical design, prototyping, and construction of VEX V5 robots, contributing to 40+ recognitions for excellence in robot design and competition performance.
- Helped scale WPRA into a top-ranked VEX Robotics academy in British Columbia, with more than 15 active teams.
- Captained teams to qualify for the VEX World Championship each season, including a 2020 Division finalist finish (top 0.4% of 20,000+ teams), while coordinating a multidisciplinary team under tight deadlines.

Volunteer Student Mentor

July 2020 – Sept 2023

- Volunteered 120+ hours at WPRA, handling logistics, maintaining accurate inventory records, and managing workshop maintenance before promotion to robotics instructor.

University of Waterloo Formula Electric (UWFE)

Chassis Team Member

Waterloo, ON

Sept 2025 – Present

- Design, prototype, and fabricate structural chassis components using CAD software, machining processes, and composite manufacturing techniques to ensure reliability and performance.
- Coordinate with leads and responsible engineers across multiple subsystems to implement performance-driven design improvements.

PROJECTS

Personal Portfolio Website

Jun 2026 – Present

Web Development

- Engineer a responsive personal portfolio from scratch using HTML, CSS, and Javascript to showcase engineering projects, utilizing Claude Code to accelerate workflow efficiency.

Engineering Design Notebook – Vex Team 886W

Jun 2024 – May 2025

Engineering Documentation

- Authored and organized a 300+ page engineering notebook detailing system architecture, design decisions, test results, and iterative improvements, ensuring clear, searchable documentation for the team.

VEX Robotics PID & Odometry System – Team 604X

Sept 2023 – May 2024

Control Systems, Programming, and Troubleshooting

- Implemented a C++ PID and odometry system to model and track the robot's position in real time, reducing steady-state error and improving path-tracking accuracy.

KEY SKILLS

Software: C++ (Basic), Python (Basic), Java (Basic), Microsoft 365, Visual Studio Code

Engineering Design: Fusion 360, Onshape, SolidWorks, Composite manufacturing, machine tools, 3D printing

Web Development: HTML (Basic), CSS (Basic), Javascript (Basic)

Languages: English (Fluent), Mandarin (Conversational), Spanish (Elementary)